



BIT1123/BISE2093/DIT1113

BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 - INDIVIDUAL

(20%)

ALL SECTIONS

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

CYBERJAYA CAMPUS

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Self-Reflective Report

3.1 Introduction

This self-reflective report summarizes my learning journey throughout the ten-week Java Object-Oriented Programming tutorial modules. The main objective of these tutorials was to understand core OOP principles and apply version control best practices using Git and GitHub.

3.2 Knowledge Gained from Each Tutorial

Through the 10-week tutorials, I acquired practical experience in core software engineering skills. Week 1 built foundational knowledge in Git workflows and basic Java output. Week 2 to Week 7 structured my understanding of object-oriented design, transitioning from simple classes to complex abstractions and inheritance hierarchies. Weeks 8–10 taught me data persistence via File I/O, dynamic data structures with ArrayList, and desktop application interface design using Java Swing.

3.3 Challenges Encountered

During the tutorials, I encountered several technical challenges:

- Git merge conflicts and syntax errors when committing repository updates.
- Understanding method overriding versus method overloading concepts in inheritance.
- Handling Java IOException during file reading/writing operations.
- Layout management and event-driven logic structuring in Java Swing GUIs.

3.4 How the Challenges Were Overcome

To overcome these obstacles, I systematically reviewed Java compiler error logs and debugged code line by line. For Git issues, I practiced commands in local terminals before pushing updates. For File I/O errors, I implemented proper try-catch exception blocks. For GUI challenges, I studied Java Swing event listeners and component bounds.

3.5 Improvements in Java Programming Skills

My programming capabilities improved significantly throughout this module. I transformed from writing simple procedural scripts into designing structured, modular, and maintainable object-oriented software applications.

3.6 Understanding of Object-Oriented Programming Concepts

I developed a comprehensive understanding of the four pillars of OOP:

- **Encapsulation:** Hiding data attributes using private modifiers and exposing controlled access via getters and setters.
- **Inheritance:** Extending parent classes to promote code reuse.
- **Polymorphism:** Redefining parent class methods inside child classes using `@Override`.
- **Abstraction:** Defining high-level contracts using abstract classes while concealing internal logic.

3.7 Future Learning Plans

In the future, I plan to expand my Java knowledge by studying advanced software architecture, exploring database connectivity (JDBC), and building full-stack web applications using the Spring Boot framework.

3.8 Conclusion

The Java OOP tutorial series provided a solid foundation in modern programming practices. Mastering Git workflows alongside OOP principles has prepared me well for higher-level software engineering assignments.

- **Advanced Java & Frameworks:** I want to study the **Spring Boot** framework to learn how to build web applications and REST APIs. I also want to learn more about Java multithreading and performance optimization.
- **Database Integration:** Instead of saving data into text files (.txt), I plan to learn **JDBC** and **MySQL/PostgreSQL** databases to manage data more securely and efficiently.
- **Software Testing & Design Patterns:** I hope to learn automated testing using **JUnit** and study common Java **Design Patterns** to write cleaner and more professional code.

3.9 GitHub Repository URL

[eiphyuphyuphway834-hue/Ei_Phyu_Phyu_Phway](https://github.com/eiphyuphyuphway834-hue/Ei_Phyu_Phyu_Phway)